

SIMATS Online Lab

CSA0925-PROGRAMMING-IN-J

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**SIMATS ENGINEERING**

**SIMATS Online Programming Lab**

JAVA PROGRAMMING

Kanigasri M 192565010RD

QUESTIONS 20 / 20 completed

Employee

**EMPLOYEE**

Write a Java program to implement a basic employee management system using inheritance. Create a parent class called Employee with attributes such as name, id, and salary. Then, create child classes such as Manager and SalesPerson that

**PROGRAM EDITOR**

Java Run Save

```
1 import java.util.*;
2 class Employee {
3     protected String name;
4     protected int id;
5     protected double salary;
6     public Employee(String name, int id, double salary) {
7         this.name = name;
8         this.id = id;
9         this.salary = salary;
10    }
11    public void displayDetails() {
12        System.out.println("Name: " + name);
13        System.out.println("ID: " + id);
14        System.out.println("Salary: " + salary);
15    }
16    public void raiseSalary(double amount) {
17        if (amount > 0) {
18            salary += amount;
19        }
20    }
21    public double getSalary() {
```

**INPUT**

Your INPUT goes here!

**OUTPUT**

Compilation Error:

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QUESTIONS 20 / 20 completed

Student-Const

**STUDENT-CONST**

Write a Java program to create a class called Student with a constructor that takes in a name, ID number, and a list of grades, and methods to calculate and return the student's average grade and letter grade.

**PROGRAM EDITOR**

Java Run Save

```
1 public class Main {
2     public static void main(String[] args) {
3         Student s = new Student("Kaniga", 192565010, 850, 1000);
4         s.display();
5     }
6 }
7 class Student {
8     String name;
9     int id;
10    int marks;
11    int maxMarks;
12    Student(String name, int id, int marks, int maxMarks) {
13        this.name = name;
14        this.id = id;
15        this.marks = marks;
16        this.maxMarks = maxMarks;
17    }
18    char calculateGrade() {
19        double per = (marks * 100.0) / maxMarks;
20        if (per >= 90) return 'A';
21        else if (per >= 80) return 'B';
```

**INPUT**

Your INPUT goes here!

**OUTPUT**

Name: Kaniga

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Partly cloudy

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ENGINEERING

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JAVA PROGRAMMING

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192565010ARD

QUESTIONS 20 / 20 completed

Bank-Const ▾

BANK-CONST

Write a Java program to create a class called BankAccount with a constructor that takes in an account number and an initial balance, and methods to deposit and withdraw money from the account.

PROGRAM EDITOR

Java ▾ Run Save

```
1 public class Main {
2     public static void main(String[] args) {
3         BankAccount account = new BankAccount("Kaniga Sri", 1000.0);
4         account.display();
5         account.deposit(500.0);
6         account.withdraw(200.0);
7         account.withdraw(2000.0);
8         account.display();
9     }
10 }
11 class BankAccount {
12     String accountHolder;
13     double balance;
14     BankAccount(String name, double initialBalance) {
15         this.accountHolder = name;
16         this.balance = initialBalance;
17     }
18     void deposit(double amount) {
19         if (amount > 0) {
20             balance += amount;
21             System.out.println("Deposited: " + amount);
22         }
23     }
24 }
```

OUTPUT

Holder: Kaniga Sri | Balance: 1000.0

INPUT

Your INPUT goes here!

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Search

L S L

WhatsApp Spotify VS Code Chrome

Taskbar icons

ENG IN 28% 08:11 PM 12-08-2026

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Search

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12-08-2026

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QUESTIONS 20 / 20 completed

Person

**PERSON**

Write a Java program to create a class called Person with a constructor that takes in a name and age, and a method to print out the person's name and age.

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main {
2     public static void main(String[] args) {
3         Person p1 = new Person("Kaniga", 20);
4         p1.display();
5     }
6 }
7 class Person {
8     String name;
9     int age;
10    Person(String name, int age) {
11        this.name = name;
12        this.age = age;
13    }
14    void display() {
15        System.out.println("Name: " + name);
16        System.out.println("Age: " + age);
17    }
18 }
```

OUTPUT

Name: Kaniga

INPUT

Your INPUT goes here!

Snipping Tool

Screenshot copied to clipboard  
Automatically saved to screenshots folder.

Mark-up and share

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QUESTIONS 20 / 20 completed

Rect-Const

**RECT-CONST**

Write a Java program to create a class called Rectangle with a constructor that takes in the length and width of the rectangle, and a method to calculate and return the area of the rectangle.

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main {
2     public static void main(String[] args) {
3         Rectangle r = new Rectangle(10, 5);
4         System.out.println("Area = " + r.area());
5     }
6 }
7 class Rectangle {
8     int length, width;
9     Rectangle(int length, int width) {
10        this.length = length;
11        this.width = width;
12    }
13    int area() {
14        return length * width;
15    }
16 }
```

OUTPUT

Area = 50

INPUT

Your INPUT goes here!

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QUESTIONS 20 / 20 completed

Overriding

OVERLOADING

Write a Java program to demonstrate method overloading with variable arguments. Create a method called sum() that takes in a variable number of integers and returns their sum. Then, create an overloaded version of the sum() method that takes in a variable

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main {
2     public static void main(String[] args) {
3         Calculator basic = new Calculator();
4         Calculator advanced = new AdvancedCalculator();
5         System.out.print("Basic Calculator: ");
6         basic.calculate(10, 5);
7         System.out.print("Advanced Calculator: ");
8         advanced.calculate(10, 5);
9     }
10 }
11 class Calculator {
12     void calculate(int a, int b) {
13         System.out.println("Result: " + (a + b));
14     }
15 }
16 class AdvancedCalculator extends Calculator {
17     void calculate(int a, int b) {
18         int sum = a + b;
19         int product = a * b;
20         System.out.println("Sum: " + sum + ", Product: " + product);
21     }
22 }
```

OUTPUT

Basic Calculator: Result: 15

INPUT

Your INPUT goes here!

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JAVA PROGRAMMING

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QUESTIONS 20 / 20 completed

Overloading

OVERLOADING

Write a Java program to demonstrate method overloading with variable arguments. Create a method called sum() that takes in a variable number of integers and returns their sum. Then, create an overloaded version of the sum() method that takes in a variable

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main {
2     public static void main(String[] args) {
3         System.out.println(sum(10, 20));
4         System.out.println(sum(1, 2, 3));
5         System.out.println(sum(1.5, 2.5));
6     }
7     static int sum(int a, int b) {
8         return a + b;
9     }
10    static int sum(int a, int b, int c) {
11        return a + b + c;
12    }
13    static double sum(double a, double b) {
14        return a + b;
15    }
16 }
```

OUTPUT

30

INPUT

Your INPUT goes here!

Snipping Tool

Screenshot copied to clipboard  
Automatically saved to screenshots folder.

Mark-up and share

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Windows taskbar showing system tray icons: weather (31°C Partly cloudy), search bar, taskbar icons (File Explorer, Edge, Word, PowerPoint, Teams, OneDrive, Outlook, Chrome, Firefox, VLC media player), system status (Bluetooth, Wi-Fi, battery level at 26%, time 08:09 PM, date 12-08-2026).

31°C Partly cloudy Search [Taskbar icons: File Explorer, Edge, Word, PowerPoint, Teams, OneDrive, Outlook, Chrome, Firefox, VLC, Spotify, Discord, Zoom, Slack, WhatsApp, Telegram, Signal, Messenger, Email, Calendar, Clock, Battery 26%, 08:09 PM 12-08-2026]

31°C Partly cloudy Search [Taskbar icons: File Explorer, Edge, L, ChatGPT, VS Code, WhatsApp, Spotify, OBS, Google, Folder] ENG IN 25% 08:08 PM 12-08-2026



Windows taskbar showing system tray icons: weather (31°C, Partly cloudy), search bar, task view button, and application icons (Edge, Word, PowerPoint, Teams, OneDrive, Outlook, Chrome, File Explorer). System status shows ENG IN, network, volume, battery (25%), and date/time (08:08 PM, 12-08-2026).

31°C Partly cloudy Search 08:07 PM 12-08-2026

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Search

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Star, Print, Close icons

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JAVA PROGRAMMING

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QUESTIONS 20 / 20 completed

Games

**GAMES**

Write a Java program to implement a basic game system using inheritance. Create a parent class called Game with attributes such as title and genre. Then, create child classes such as ActionGame and PuzzleGame that inherit from the Game class and have their

PROGRAM EDITOR

Java Run Save

```
1 public class Main {
2     public static void main(String[] args) {
3         Player p1 = new Player("Hero", 100);
4         Enemy e1 = new Enemy("Monster", 50);
5         p1.attack(e1);
6         e1.attack(p1);
7         p1.displayStatus();
8         e1.displayStatus();
9     }
10 }
11 class Entity {
12     String name;
13     int health;
14
15     Entity(String name, int health) {
16         this.name = name;
17         this.health = health;
18     }
19     void displayStatus() {
20         System.out.println(name + " Health: " + health);
21     }
22 }
```

OUTPUT

Hero attacks Monster!

INPUT

Your INPUT goes here!

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Star, Print, Close icons

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QUESTIONS 20 / 20 completed

University

**UNIVERSITY**

Write a Java program to implement a basic university management system using inheritance. Create a parent class called Person with attributes such as name and age. Then, create child classes such as Student and Professor that inherit from the Person class

PROGRAM EDITOR

Java Run Save

```
1 public class Main {
2     public static void main(String[] args) {
3         Student s = new Student("Kaniga", 1);
4         s.display();
5         Teacher t = new Teacher("Mr. John", "CSE");
6         t.display();
7     }
8 }
9 class UniversityMember {
10     String name;
11     UniversityMember(String name) { this.name = name; }
12     void display() { System.out.println("Name: " + name); }
13 }
14 class Student extends UniversityMember {
15     int id;
16     Student(String name, int id) {
17         super(name);
18         this.id = id;
19     }
20     void display() {
21         super.display();
22     }
23 }
```

OUTPUT

Name: Kaniga

INPUT

Your INPUT goes here!



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QUESTIONS 20 / 20 completed

Library

LIBRARY

Write a Java program to implement a basic library management system using inheritance. Create a parent class called Item with attributes such as title and author. Then, create child classes such as Book and DVD that inherit from the Item class and have their

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main {
2     public static void main(String[] args) {
3         Book b = new Book("Java", "R1");
4         b.display();
5
6         Magazine m = new Magazine("Tech Today", 10);
7         m.display();
8     }
9 }
10 class LibraryItem {
11     String title;
12     LibraryItem(String title) { this.title = title; }
13     void display() { System.out.println("Title: " + title); }
14 }
15 class Book extends LibraryItem {
16     String author;
17     Book(String title, String author) {
18         super(title);
19         this.author = author;
20     }
21     @Override
```

OUTPUT

Title: Java

INPUT

Your INPUT goes here!

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QUESTIONS 20 / 20 completed

Shapes-1

SHAPES-1

Write a Java program to implement a basic shape hierarchy using inheritance. Create a parent class called Shape with attributes such as color and area. Then, create child classes such as Circle, Square, and Triangle that inherit from the Shape class and have their own unique

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main {
2     public static void main(String[] args) {
3         Shape redSq = new Square("Red", 5);
4         Shape blueTri = new Triangle("Blue", 4);
5         Shape greenCir = new Circle("Green", 3);
6         redSq.draw();
7         System.out.println(redSq.area());
8         blueTri.draw();
9         System.out.println(blueTri.area());
10        greenCir.draw();
11        System.out.println(greenCir.area());
12    }
13 }
14 abstract class Shape {
15     String color;
16     Shape(String color) { this.color = color; }
17     abstract double area();
18     void draw() { System.out.println(color + " " + getClass().getSimpleName()); }
19 }
20 class Square extends Shape {
21     double side;
```

OUTPUT

Red Square

INPUT

Your INPUT goes here!

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QUESTIONS 20 / 20 completed

Animal-1

**ANIMAL-1**

Write a Java program to implement a basic animal classification system using inheritance. Create a parent class called Animal with attributes such as name and habitat. Then, create child classes such as Mammal, Reptile, and Bird that inherit from the Animal class and

**PROGRAM EDITOR**

Java Run Save

```
1 public class Main {
2     public static void main(String[] args) {
3         Animal a1 = new Dog("Rex");
4         a1.eat();
5
6         Animal a2 = new Cat("Kitty");
7         a2.eat();
8
9         Animal a3 = new Cow("Moo");
10        a3.eat();
11    }
12 }
13 abstract class Animal {
14     String name;
15     Animal(String name) { this.name = name; }
16     abstract void eat();
17 }
18 class Dog extends Animal {
19     Dog(String name) { super(name); }
20     void eat() { System.out.println(name + " is a Carnivore"); }
21 }
```

**OUTPUT**

Rex is a Carnivore

**INPUT**

Your INPUT goes here!

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QUESTIONS 20 / 20 completed

Employee

**EMPLOYEE**

Write a Java program to implement a basic employee management system using inheritance. Create a parent class called Employee with attributes such as name, id, and salary. Then, create child classes such as Manager and SalesPerson that

**PROGRAM EDITOR**

Java Run Save

```
1 import java.util.*;
2 class Employee {
3     protected String name;
4     protected int id;
5     protected double salary;
6     public Employee(String name, int id, double salary) {
7         this.name = name;
8         this.id = id;
9         this.salary = salary;
10    }
11    public void displayDetails() {
12        System.out.println("Name: " + name);
13        System.out.println("ID: " + id);
14        System.out.println("Salary: " + salary);
15    }
16    public void raiseSalary(double amount) {
17        if (amount > 0) {
18            salary += amount;
19        }
20    }
21    public double getSalary() {
```

**OUTPUT**

Compilation Error:

**INPUT**

Your INPUT goes here!

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JAVA PROGRAMMING

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1923650101010

QUESTIONS 20 / 20 completed

Bank

**BANK**

Write a Java program to implement a basic banking system using inheritance. Create a parent class called Account with attributes such as account number, account holder name, and balance. Then, create child classes such as SavingsAccount and

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main {
2     public static void main(String[] args) {
3         SavingsAccount s = new SavingsAccount("Kaniga", 5000, 4.5);
4         s.deposit(1000);
5         s.withdraw(2000);
6         s.applyInterest();
7         s.display();
8     }
9 }
10 class Account {
11     String name;
12     double balance;
13     Account(String name, double balance) {
14         this.name = name;
15         this.balance = balance;
16     }
17     void deposit(double amt) {
18         balance += amt;
19     }
20     void withdraw(double amt) {
21         if (amt <= balance) balance -= amt;
```

OUTPUT

Name: Kaniga, Balance: 4180.0

INPUT

Your INPUT goes here!

Rain warning  
In effect

Search

ENG IN

22%

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12-08-2026